

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2023/24 Semester 2

MH1301 Discrete Mathematics

Tutorial 5

For the tutorial in Week 6 on 22 February, let us discuss

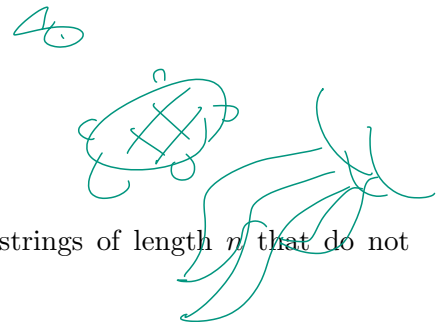
- Ex. 8.1.9, 8.1.11, 8.2.2(a)–(e), 8.2.5, 8.2.6, additional exercises.

Additional Exercise on tiling. You need to tile a $n \times 1$ hallway with unlimited supply of 1×1 red tiles, 2×1 red tiles, and 2×1 blue tiles. Write down the recurrence formula and initial conditions for the number of ways to tile:

- (a) an $n \times 1$ hallway for which you are allowed to use all types of tiles.
- (b) an $n \times 1$ hallway for which you are allowed to use all types of tiles, but no two red tiles are consecutive.

Ex. 8.1.9. A ternary string is a string containing only 0's, 1's, and 2's.

- (a) Find a recurrence relation for the number of ternary strings of length n that do not contain two consecutive 0's.
- (b) What are the initial conditions ?
- (c) Find the answer when $n = 4$.



Ex. 8.1.11.

- (a) Find a recurrence relation for the number of ternary strings of length n that do not contain two consecutive 0's or two consecutive 1's.
- (b) What are the initial conditions ?
- (c) Find the answer when $n = 4$.

Ex. 8.2.2 (a)–(c). Solve the following homogeneous recurrence relations.

- (a) $a_n = 2a_{n-1}$ for $n \geq 1, a_0 = 3$.
- (b) $a_n = a_{n-1}$ for $n \geq 1, a_0 = 2$.
- (c) $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2, a_0 = 1, a_1 = 0$.

Ex. 8.2.2 (d), (e). Solve the following homogeneous recurrence relations.

- (d) $a_n = 4a_{n-1} - 4a_{n-2}$ for $n \geq 2, a_0 = 6, a_1 = 8$.
- (e) $a_n = -4a_{n-1} - 4a_{n-2}$ for $n \geq 2, a_0 = 0, a_1 = 1$.

Ex. 8.2.5. Let a_n be the number of ways to tile a $2 \times n$ rectangular board using 1×2 and 2×2 pieces.

- (a) Calculate the values a_1 and a_2 .
- (b) Find a recurrence relation for the sequence (a_n) .
- (c) Solve the recurrence relation in part (b).

Ex. 8.2.6. A model for the number of lobsters caught per year is based on the assumption that the number of lobsters caught in a year is the average of the number caught in the two previous years.

- (a) Find a recurrence relation for the sequence (L_n) , where L_n is the number of lobsters caught in year n .
- (b) Solve L_n if 100,000 lobsters were caught in year 1 and 300,000 lobsters were caught in year 2.

Additional Exercise. Solve the simultaneous recurrence relations

$$a_n = 3a_{n-1} + 2b_{n-1},$$

$$b_n = a_{n-1} + 2b_{n-1},$$

with $a_0 = 1$ and $b_0 = 2$.



Hints.

Additional Exercise on tiling.

- (a) Instead of considering the last bit, consider the last tile, is it a 1×1 red tile, or 2×1 red tile, or 2×1 blue tile?
- (b) If the last tile is a red tile, then the penultimate tile must be a blue tile.

Ex. 8.1.9. Similar to 8.1.10(a), consider the last bit, or the last two bits if necessary.

Ex 8.1.11. Let a_n be the number of ternary strings of length n that do not contain two consecutive 0's or two consecutive 1's. Define $a_{0,n}$ to be the number of ternary strings of length n that do not contain two consecutive 0's or two consecutive 1's and **ends with 0**. Define $a_{1,n}$ and $a_{2,n}$ similarly. Then clearly

$$a_n = a_{0,n} + a_{1,n} + a_{2,n}.$$

Express $a_{0,n}$ in terms of $a_{0,n-1}$, $a_{1,n-1}$, and $a_{2,n-1}$. Do likewise for $a_{1,n}$, and $a_{2,n}$.

Ex 8.2.5 As usual, consider the last column of the board. Note that a 1×2 piece can be placed horizontally or vertically.